

Yeswanth Chamarthy

Tampa, FL • chamarthyeswanth@gmail.com • LinkedIn • GitHub • (813)-593-1789

PROFESSIONAL EXPERIENCE

McKibbon Hospitality

Senior Business Analyst | Data Engineering & GenAI

Tampa, FL

June 2024 – Present

As a **Data & AI Engineer** at McKibbon Hospitality, I built **cloud-scalable ETL pipelines** and deployed **GenAI-driven workflows** that accelerated insights delivery, improved data reliability, and enabled self-serve access to trusted information across the organization.

- ❖ **Text-to-SQL Bot for Financial Analytics (LangChain, GPT-4o-mini, AWS Bedrock (Claude 3 Haiku), DynamoDB, AWS Athena, AWS Lambda, SQS, Python, Boto3, Bot Framework, Azure Bot Services)**
 - Built a Teams-based **multimodal Text-to-SQL** bot that generates Athena queries from natural language and returns concise business summaries.
 - Orchestrated a **LangChain multi-step workflow** to route tasks between models; utilized GPT-4o-mini for precise SQL generation and Claude 3 Haiku (Bedrock) for natural language summarization.
 - Added SQL guardrails (SELECT-only, table/column allowlist, required filters) to improve query accuracy and prevent expensive scans.
 - Designed an event-driven processing system using **SQS and Lambda** to handle high-throughput requests.
 - Implemented **DynamoDB result caching** to reuse query outputs for repeat questions and reduce repeated Athena executions.
 - Built **LangChain Tool** interface (**MCP**) to expose secure, reusable tools (Athena query execution, schema lookup, caching) so the bot can retrieve data through a controlled interface.
 - **Impact: Observed ~18-30% cache hit rate; reduced response time for repeat queries by ~40–70%**
- ❖ **RAG Knowledge Assistant for HR department (LangChain, AWS Bedrock (Nova Lite, Titan Text Embeddings), ChromaDB, Python, Boto3, Bot Framework, AWS Lambda, S3, GIT, Azure Bot Services, Azure App Service)**
 - Built a **Retrieval-Augmented Generation (RAG)** assistant to answer HR policies and forms, deployed through a **Microsoft Teams bot** for self-serve access.
 - Engineered a **LangChain-based ingestion pipeline** utilizing recursive character text splitters for chunking and Titan embeddings for vectorization in ChromaDB.
 - Integrated **AWS Bedrock Titan Text Embeddings v2** for retrieval and **Bedrock Nova Lite**.
 - Implemented **LangChain's ConversationBufferMemory** to maintain context across multi-turn user interactions, reducing hallucinations and improving follow-up accuracy.
 - **Impact: ~0.98 faithfulness score in evaluation; indexed ~7K+ chunks for HR document search.**
- ❖ **Agentic RAG & Automated Document Processor (React, Typescript, LangChain, AWS Bedrock (Claude 3.5 Sonnet), Titan Embeddings, Python, OCR, SharePoint API)**
 - Developed an **Agentic RAG system** to automate the lifecycle of legal Contracts, Licenses, and Permits reducing manual data entry, featuring a responsive **React and TypeScript** dashboard for real-time document tracking.
 - **Engineered a type-safe frontend interface** using **TypeScript** to visualize the autonomous extraction process, allowing users to review and validate JSON-structured data before it synchronizes with **SharePoint Lists**.
 - Engineered a **fail-safe ingestion engine** using a multi-modal approach (PyMuPDF, OCR) to ensure 100% text extraction accuracy from low-quality scans.
 - **Architected an Autonomous Agent** (Claude 3.5 Sonnet) for zero-shot data extraction; integrated the backend with the React frontend via REST APIs to deliver a seamless, end-to-end user experience.
 - Optimized retrieval using **Titan Embeddings v2**, achieving high-precision semantic search across a repository of 200 documents.
- ❖ **BI Platform Migration & Data Pipeline Modernization (Qlik Sense, Tableau, AWS Lambda, Glue, Athena, Airflow, Step Functions, S3, EC2, Delta Lake, REST API, Datadog)**
 - Led the end-to-end migration from **Qlik sense** to **Tableau** by designing and implementing AWS- native ETL pipelines, analytics views, and Tableau-ready data models.
 - Built **50+ Lambda functions, 20+ AWS Glue Spark ETL Jobs and 10+ Step Functions** to automate data ingestion, transformation, and downstream refresh workflows.
 - Developed multiple **REST API integrations** for data collection from external/internal sources and standardized ingestion patterns for reliability and reusability.
 - Orchestrated pipelines and dependencies using **Apache Airflow**, enabling scheduled and event-based processing across Lambda, Glue, and API workloads.

- Implemented a **medallion architecture** on **S3**, created **50+ Athena views** for analytical processing on Delta Lake, and used Athena views as Tableau sources for modeling.
- Migrated and consolidated **1TB+ data** from diverse sources into **PostgreSQL**, improving accessibility for downstream analytics and operational use cases.
- Established monitoring and alerting using **Datadog** to track job health, failures, latency, and SLA adherence across the pipeline ecosystem.
- Delivered **18 Tableau dashboards** adopted across departments, supporting standardized reporting and reducing manual reporting overhead.
- **Impact:** Saved \$100K/year by transitioning from Qlik Sense to Tableau and reduced data load time by ~40% through platform modernization, improving reliability and scalability of org-wide reporting.

University of South Florida

Research Assistant

Tampa, FL

Feb 2023 – May 2024

- Automated the report generation process within **MySQL** by implementing an **ETL pipeline** using **Apache Airflow**, resulting in increased efficiency, and **saving 1.5 hours daily**.
- Tuned TSQL queries and stored procedures in MS SQL Server, resulting in 20% performance gains.
- Formulated statistical models like multiple regression, **Tobit**, and **logistic regression** in **R** and **Python** to track restaurant customer tipping behaviour, resulting in a **revenue increase of 60k USD**.
- Visualized **Tableau** dashboards to monitor student enrolment in hospitality courses.

Tata Consultancy Services

Data Engineer

Chennai, India

April 2019 – Aug 2022

- Engineered **8+ data ETL pipelines** using **Lambda**, **Glue**, **EMR**, and step functions to migrate/store data across various sources like **S3**, **Athena**, and **DynamoDB**.
- Boosted app retention **by 6%** by developing / modifying KPIs that measure customer engagement using **PowerBI** to track user actions and usage patterns.
- Migrated Database from Microsoft SQL Server 2005 to Microsoft SQL Server 2019 using ETL.
- Managed tickets in **Jira** and made documentation with confluence for every procedure we migrated.
- Integrated various data sources and sinks in DataStage jobs, Enhanced code to leverage **PySpark's** extensive data source support (e.g., CSV, Parquet, Avro, JDBC) for reading and writing data, leading to a 40% data integration time.
- Accelerated query performance **by 20%** through strategic crafting of indexes, stored procedures in **T SQL**, and executing complex queries in **MS SQL**, minimizing joins and sub-queries for data retrieval and manipulation.
- Deployed Databricks jobs using **PySpark** to streamline and automate the daily data transfer tasks and produce Quicksight reports for consumption by various stakeholders.
- Followed Agile methodologies and visualized the team performance every month using Tableau and Excel.

EDUCATION

University of South Florida

Master of Science in Business Analytics and Information Systems

GPA: 3.93

Tampa, FL

Aug 2022 – May 2024

Gitam University

Bachelor of Technology in Mechanical Engineering

GPA: 3.81

India

June 2015 – April 2019

Skills

- **Programming Competency:** Python, SQL (MS SQL, MYSQL, Postgres), R, Shell, Unix/Bash, HTML, JavaScript, Scala
- **Generative AI:** AWS Bedrock, OpenAI API, LangChain, ChromaDB, RAG, Prompt Engineering, Agents (MCP), CrewAI
- **Data Engineering:** AWS-(EC2, S3, Lambda, Glue, CloudFormation, Redshift, EMR, SQS, Bedrock, Step Function, IAM), Apache Hadoop, PySpark, Airflow, SSIS, SSRS, HDFS, Hive, Azure App Service, Azure Bot Service
- **Visualization Tools, Libraries:** PowerBI, Tableau, Qlik Sense, Google Looker Studio, Microsoft Excel, GIT, Pandas, Scikit-Learn, Matplotlib, Seaborn, OpenCV, PowerApps, DAX
- **Methods:** Data Modeling, Data Analysis, Regression Analysis, Hypothesis Testing, Deep Learning, Agile (Scrum/Kanban), Jira